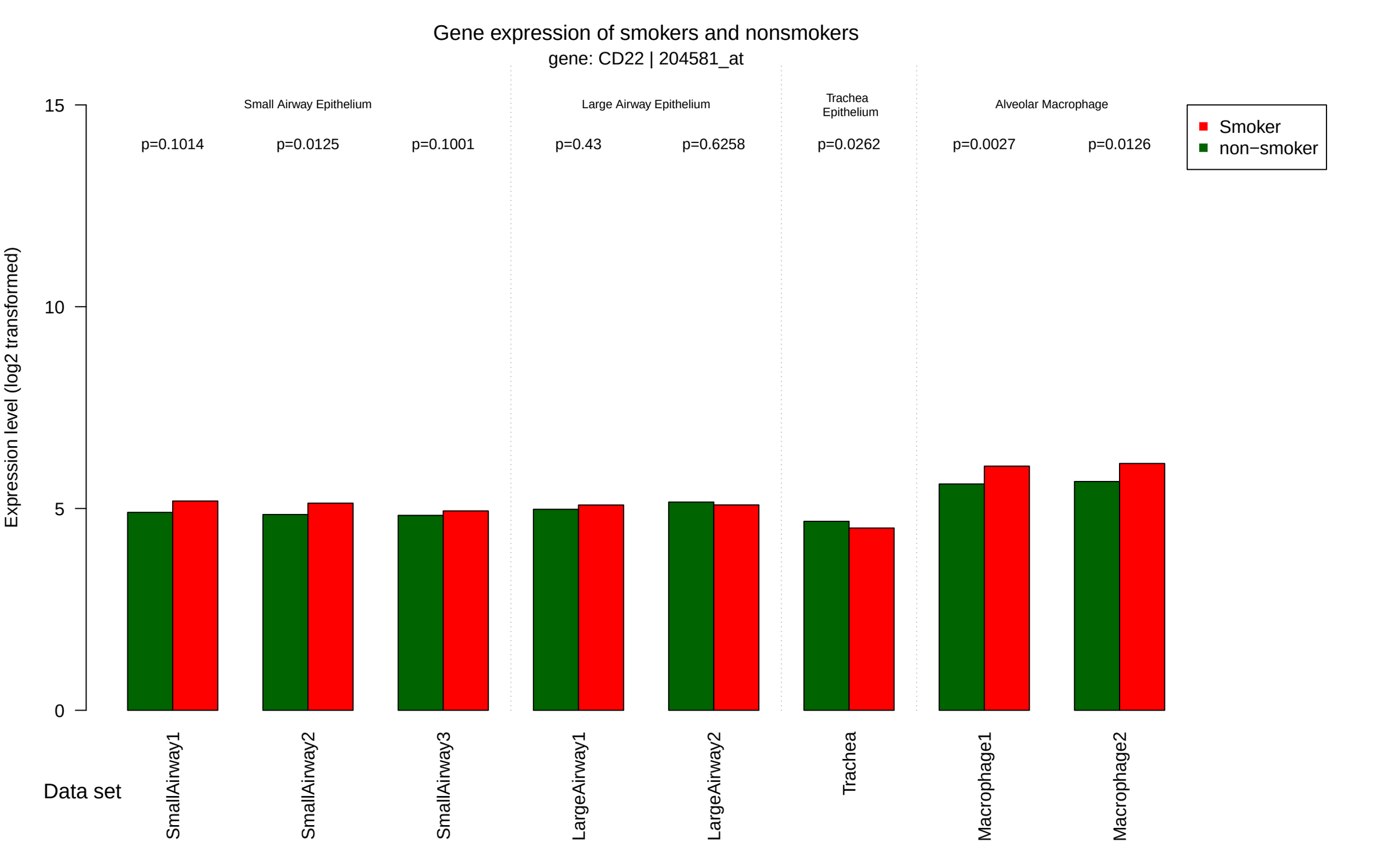
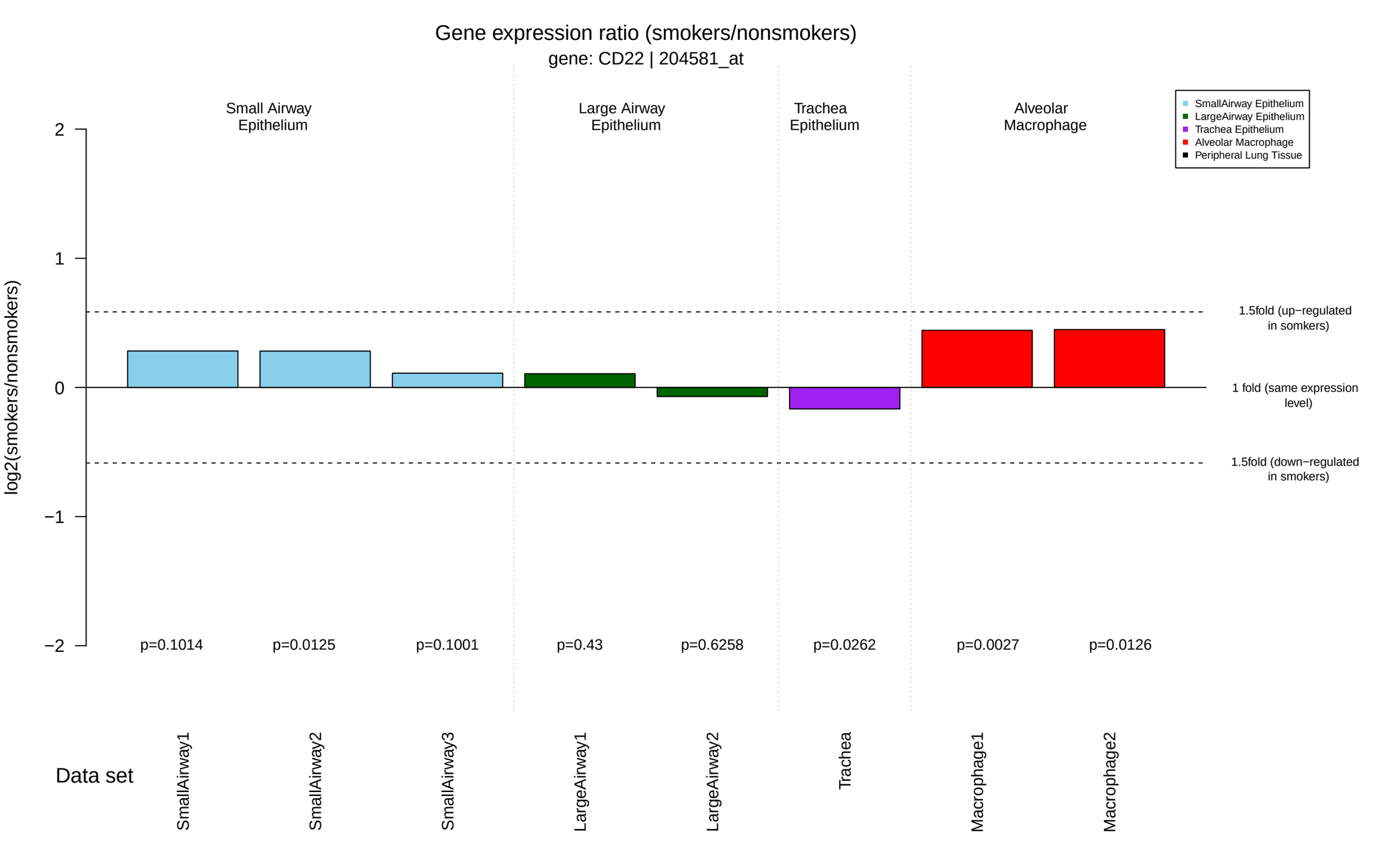
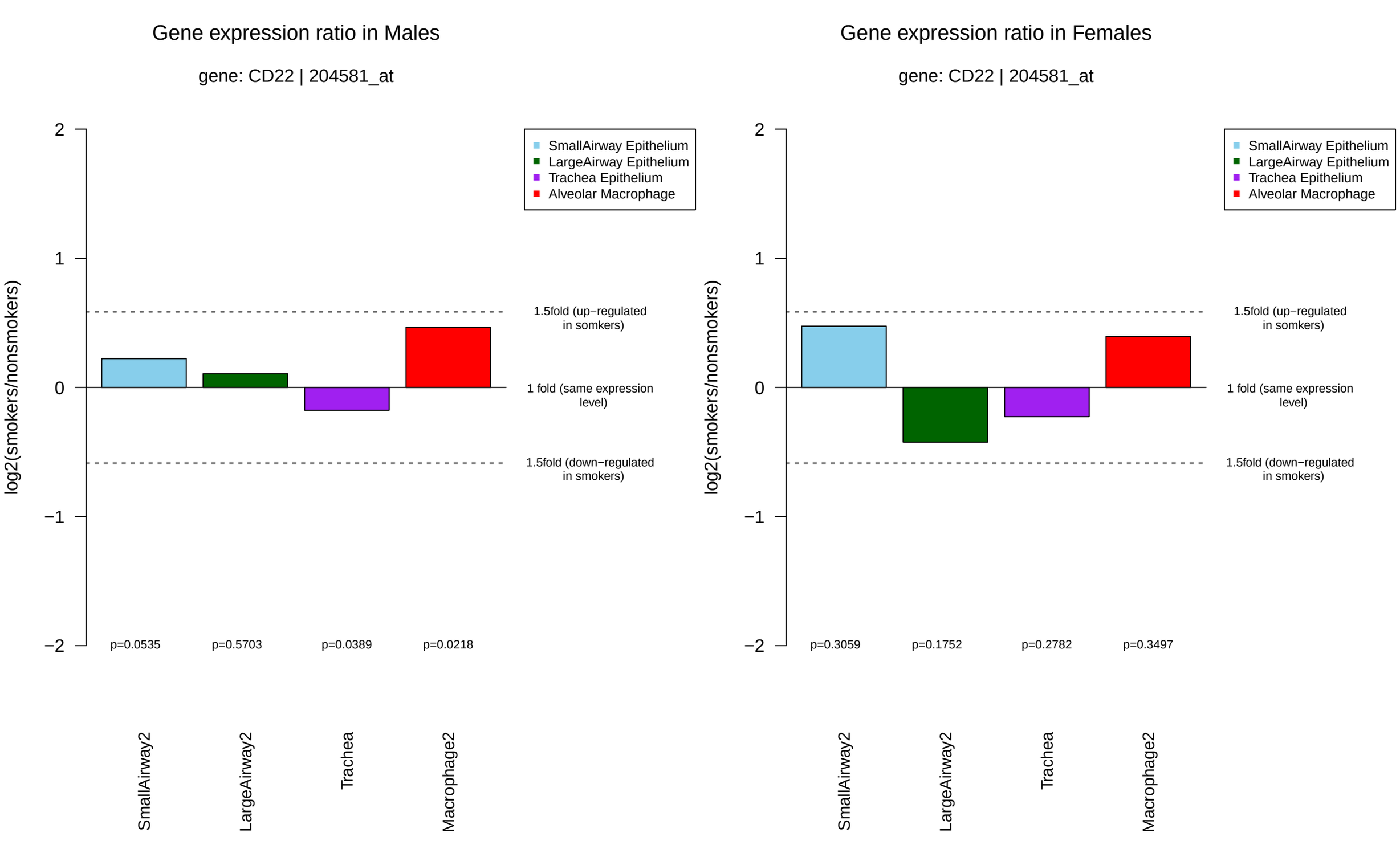
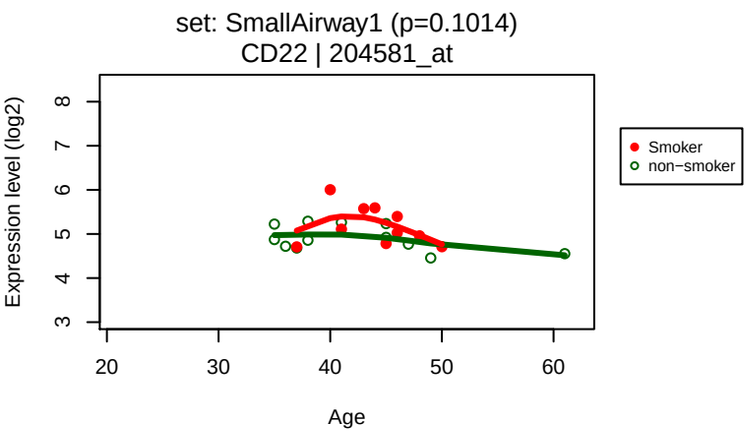
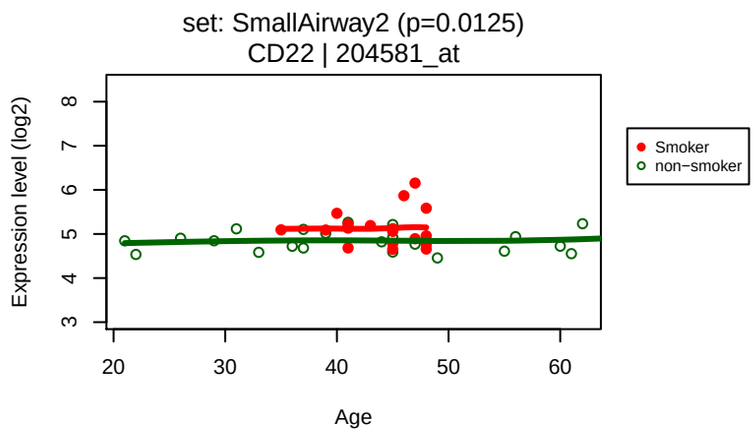
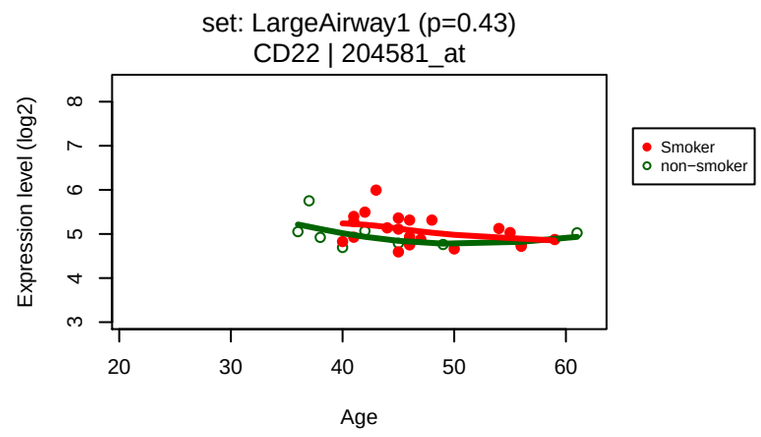

Gene expression of smokers and nonsmokers
gene: CD22 | 204581_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1014
p=0.0125
p=0.1001
p=0.43
p=0.6258
p=0.0262
p=0.0027
p=0.0126
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

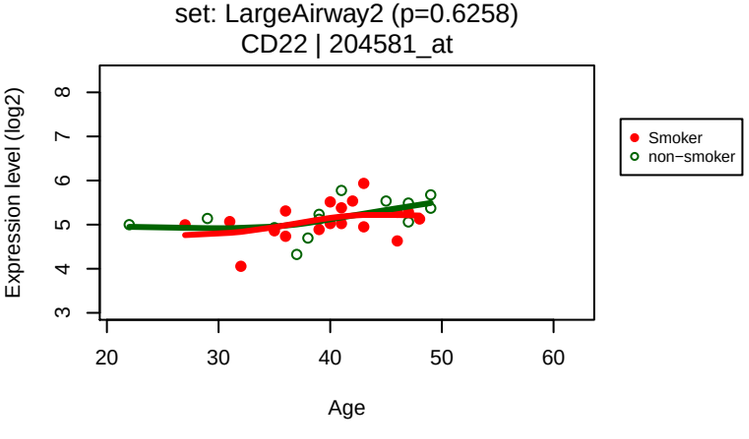

Gene expression ratio (smokers/nonsmokers)
gene: CD22 | 204581_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1014
p=0.0125
p=0.1001
p=0.43
p=0.6258
p=0.0262
p=0.0027
p=0.0126
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

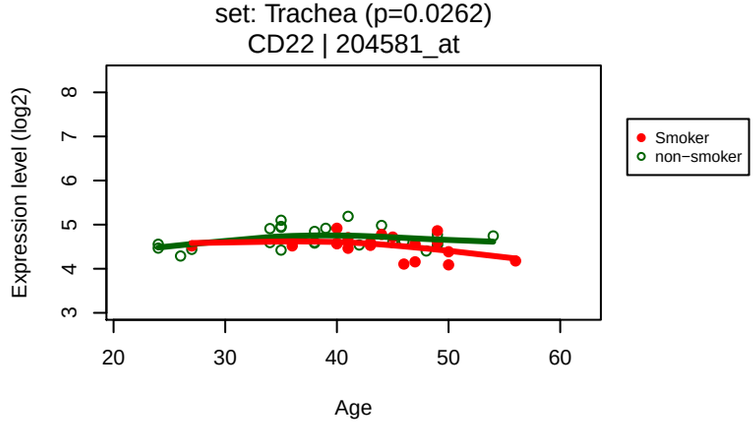

Gene expression ratio in Males
gene: CD22 | 204581_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0535
p=0.5703
p=0.0389
p=0.0218
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CD22 | 204581_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3059
p=0.1752
p=0.2782
p=0.3497
SmallAirway2
LargeAirway2
Trachea
Macrophage2

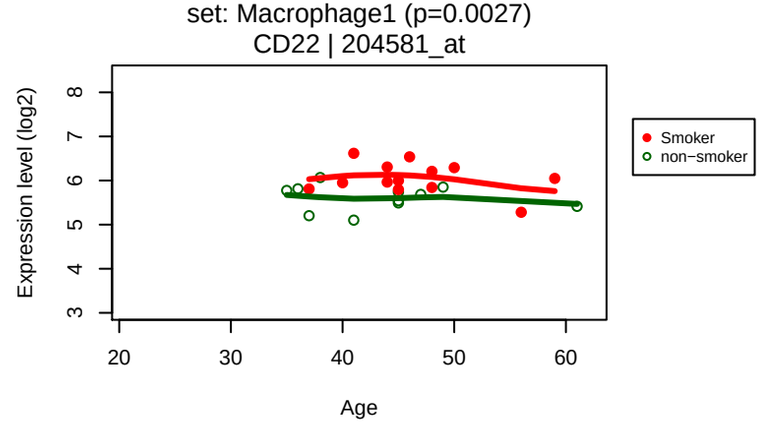

set: SmallAirway1 (p=0.1014)
CD22 | 204581_at
Expression level (log2)
Age
Smoker
non−smoker

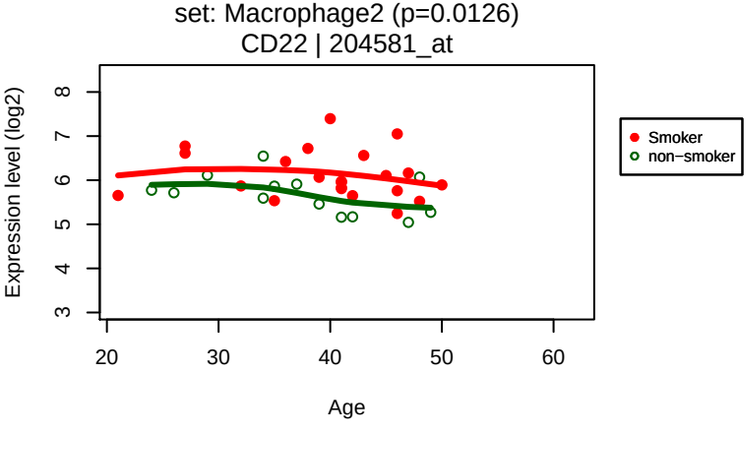

set: SmallAirway2 (p=0.0125)
CD22 | 204581_at
Expression level (log2)
Age
Smoker
non−smoker

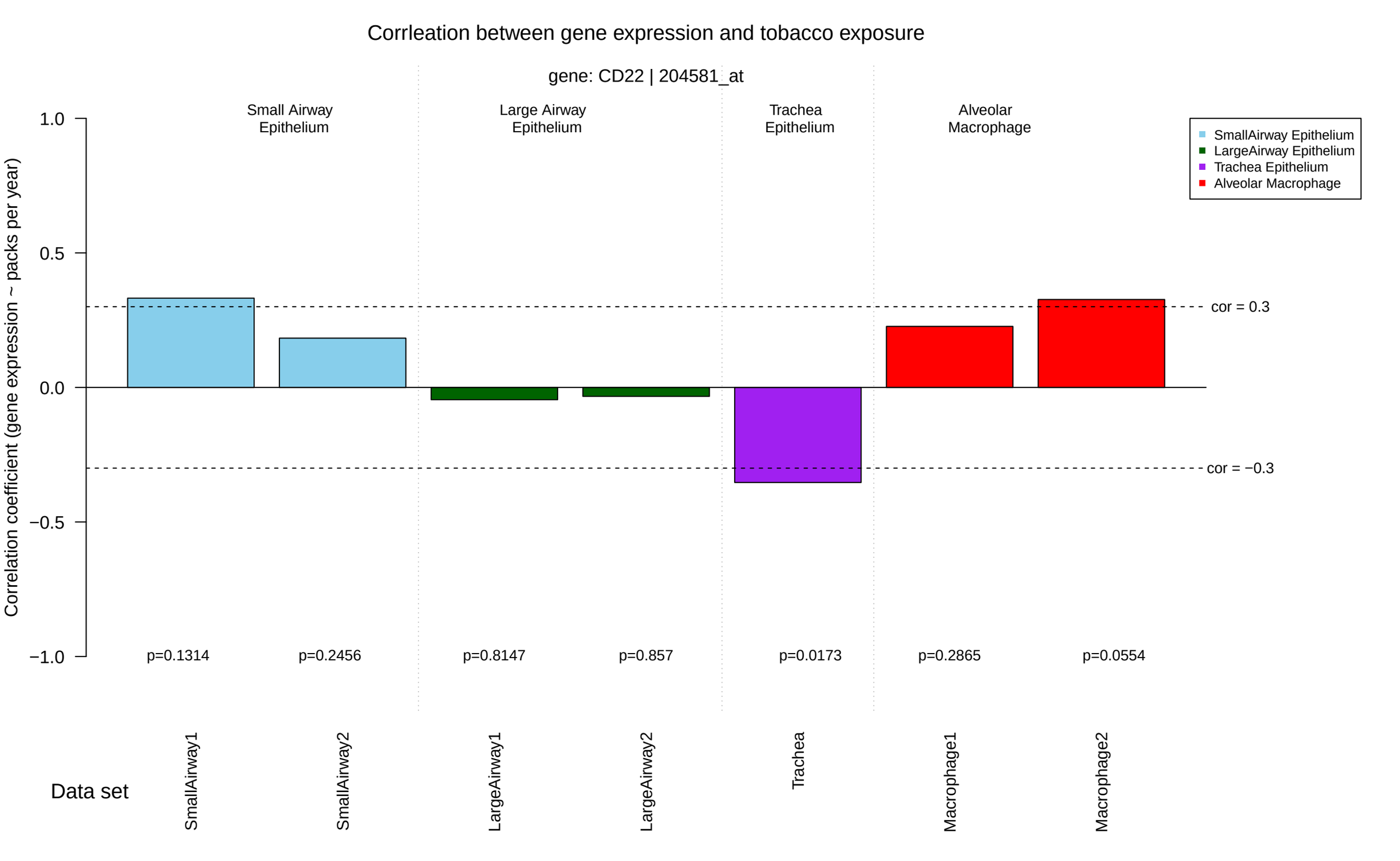

set: LargeAirway1 (p=0.43)
CD22 | 204581_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6258)
CD22 | 204581_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0262)
CD22 | 204581_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0027)
CD22 | 204581_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0126)
CD22 | 204581_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CD22 | 204581_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1314
p=0.2456
p=0.8147
p=0.857
p=0.0173
p=0.2865
p=0.0554
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2